

Challenge Write-Up Template

Instructions: Copy this template for every challenge you submit. Fill in every section — a write-up that only shows the final flag without explanation will not receive full credit. Delete these italicized instruction lines before submitting.

Challenge Information

Student Name	Facon, Janna Mae L.
Section / Class	BSIT- 3G
Challenge Name	SUDO MAKE ME A SANDWICH
Category	General Skills
Points / Difficulty	Easy
Date Solved	8/12/2026
Flag Format Given	PicoCTF{...}

1. Challenge Description

Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copypaste the original prompt.

The challenge gave us access to a remote Linux machine through SSH. Our goal was to find the hidden flag inside a file called flag.txt and submit it to the PicoCTF website. However, the file was protected and could not be read by the normal user. We needed to investigate the permissions on the file and determine whether we had any special privileges that could help us access it.

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

After connecting to the challenge machine, I first checked the contents of the current directory and found a file named flag.txt. I then checked its permissions and ownership and noticed that the file belonged to the root user, which explained why

I could not access its contents normally. When I attempted to read the file, the system returned a permission denied message. Because the file was owned by root, I suspected that the challenge involved Linux permissions and privilege escalation. I then checked the user's available sudo privileges and discovered that the ctf-player account was allowed to run /bin/emacs flag.txt as root without requiring a password. This was the important clue because Emacs can execute shell commands while running with the privileges of the user who launched it.

3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

Tool / Command	What it was used for	Result / Output
<code>ls -la</code>	To inspect the files, ownership, and permissions	Found <code>flag.txt</code> , which was owned by <code>root</code>
<code>cat flag.txt</code>	To try reading the flag normally	Returned <code>Permission denied</code>
<code>sudo -l</code>	To check available sudo privileges	Allowed <code>/bin/emacs flag.txt</code> to run as root without a password

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

Step 1

The first step that I did was read the challenge description carefully to understand what the challenge was asking me to find and what I needed to accomplish. From the description, I understood that I needed to find the flag from the protected `flag.txt` file. After understanding the objective, I proceeded to inspect the challenge environment and look for a possible way to access the file.

General Skills **Easy**

by Darkraicg492 · 📅 picoCTF 2026

Can you read the flag? I think you can!

INSTANCE

Launch Instance

Additional details appear once you launch your instance.

HINTS ⓘ

1 Reveal hint

2  Reveal hint

Step 2

The second step was to try reading the file directly using `cat flag.txt`. However, the system returned a **“Permission denied”** message. This confirmed that the current user did not have enough permissions to read the file, so I needed to find another way to access it.

[illegible]

Step 3

The third step was to check whether the current user had any special sudo privileges. I used `sudo -l` to see which commands could be executed with elevated privileges. The output showed that the `ctf-player` user was allowed to run `/bin/emacs flag.txt` as root without requiring a password. This was the main clue because running Emacs as root could potentially provide access to the protected file.

[illegible]

Step 4

The fourth step was to use the sudo permission that I discovered. I ran `sudo /bin/emacs flag.txt`, which opened the file using Emacs with root privileges. Since Emacs was running as root, I could use its shell-command functionality to interact with the system with those elevated privileges.

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The fifth step was to use the shell-command functionality inside Emacs to read the contents of `flag.txt`. Since the Emacs process had root privileges, it was able to access the file even though the normal `ctf-player` account could not. The contents of the file revealed the PicoCTF flag: `picoCTF{ju57 5ud0 17 cce7a3f7}`.

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

The main concept I learned from this challenge is **sudo privilege misconfiguration**, where a user is given permission to run a powerful program as root. Even though I could not directly read the protected flag.txt, the ability to run Emacs with root privileges allowed me to access the file. This shows why sudo permissions should be carefully restricted and only granted to trusted commands that cannot be abused for privilege escalation.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

<https://www.youtube.com/watch?v=dJwoUj9p--E>
